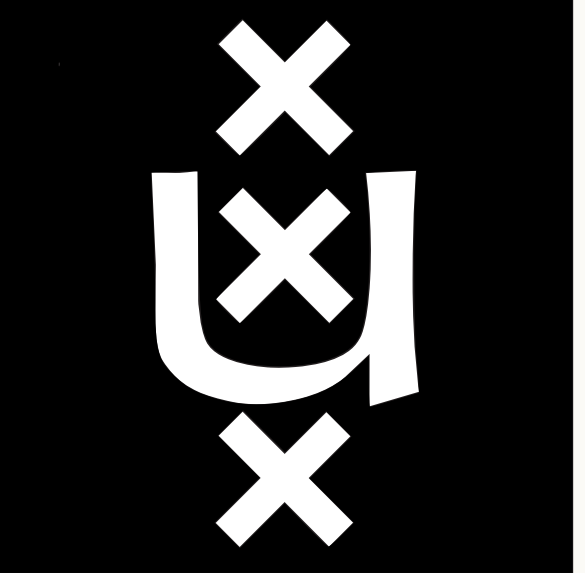




Spotting hidden financial crime patterns in networks with AI

Work package 2.1 · AI for finding crime and criminal network patterns in financial and other big data

Yassin Mohamadi, Zeno Geradts, Marcel Worring

Our goal

Develop AI models that uncover hidden patterns of crime in vast amounts of financial data—reliably and transparently.

1 Why is crime so hard for AI to find?



Hidden

Real crime data is confidential and rarely shared publicly.



Unlabelled

Nobody has marked which transactions are criminal.



Rare

Crime is a tiny fraction of all activity, so there are few examples to learn from.



Private

Personal details must be protected, and strict rules limit data sharing.



Varied

Money laundering and human trafficking leave very different traces.



No black boxes

AI model results should be transparent and provide a degree of explainability.

2 Our approach: finding what doesn't fit in a network

Learning from examples

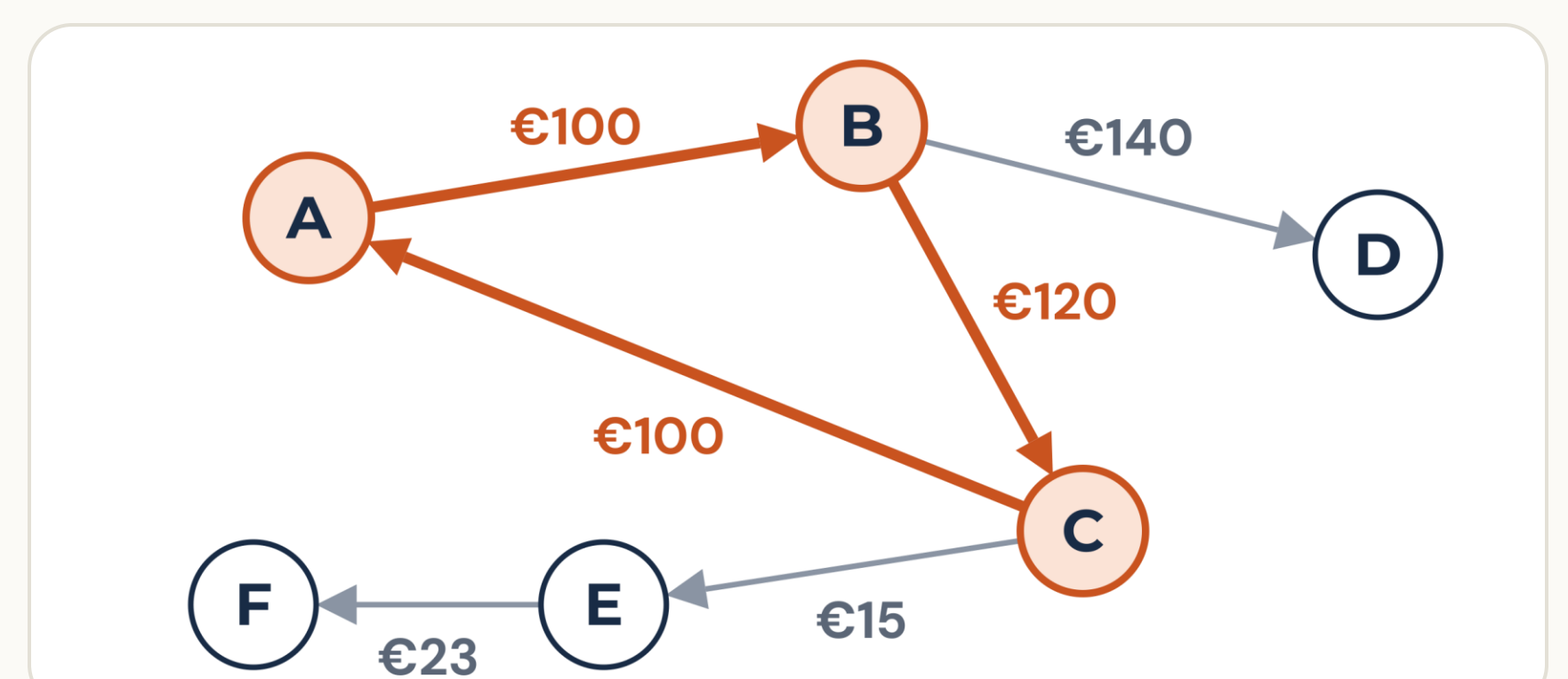
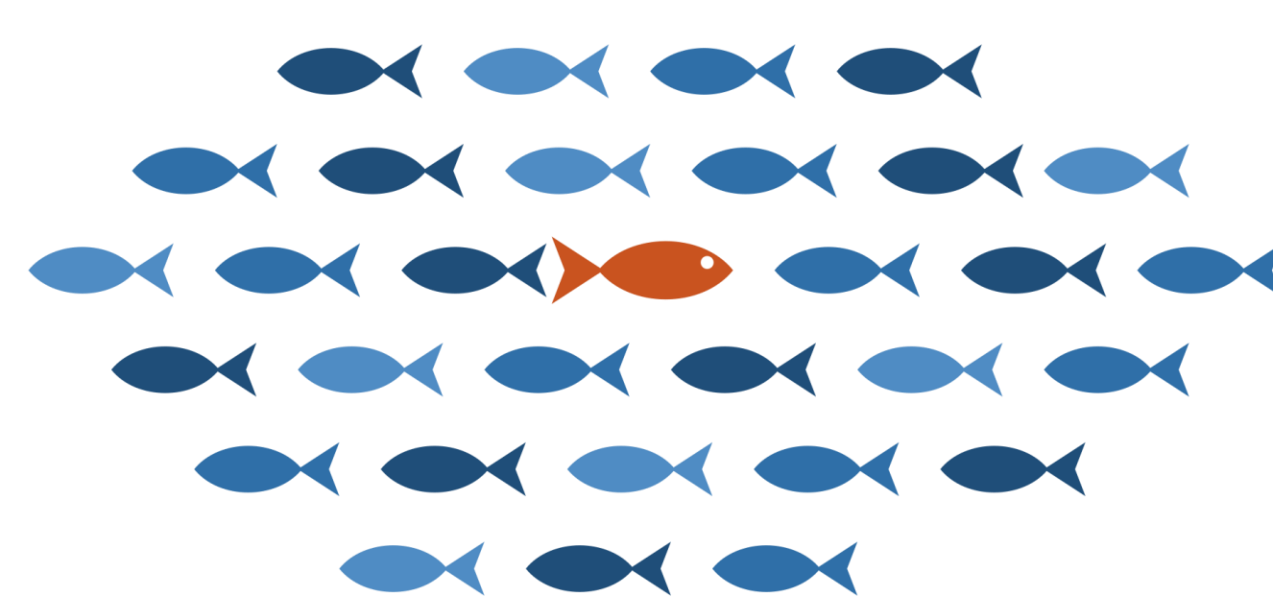


✗ needs labels

Finding groups alone



✓ no labels needed



Learn without labels

Most AI learns from examples someone has already labelled. We have none, so our AI finds the patterns on its own.

Spot the odd one out

Most financial activity is normal. The AI learns that normal pattern, so behaviour that doesn't fit stands out.

Follow the money

We study who pays whom, not single payments. Money that goes round in a circle ($A \rightarrow B \rightarrow C \rightarrow A$) is a warning sign.

3 How it works

1

Financial records

Bank transfers and other transaction data.



2

Draw a money map

Accounts become nodes; payments become links between them.



3

AI learns what's normal

It teaches itself from the data (self-supervised learning), with no crime labels needed.



4a

Unusual payments

Odd transactions, tracked over time → a risk score for each payment.

4b

Unusual accounts

Behaviour that drifts over the long term → a risk score for each account.

Our first step:



Generative AI? Real crime data is scarce, so it creates realistic new data to learn from — and, like the rest of our approach, it needs no labels to work.



The outcome: Train AI models on generated data and evaluate them on real-world data.

